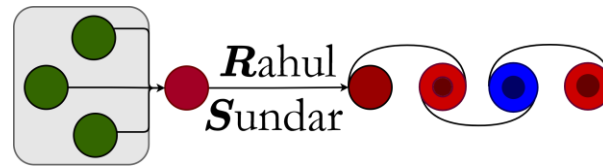


FROM NOISE TO KNOWLEDGE!

The Magic of Diffusion Models



PhD Scholar / Scientist (AI/ML)
Dept. of Aerospace Engineering, IIT
Madras / **Verisk, India**



Founder, **Dhyuti**
An open source community initiative
for AI in Science and Engineering

@Ankura AI Meets – 25th January, 2026

Your key takeaways

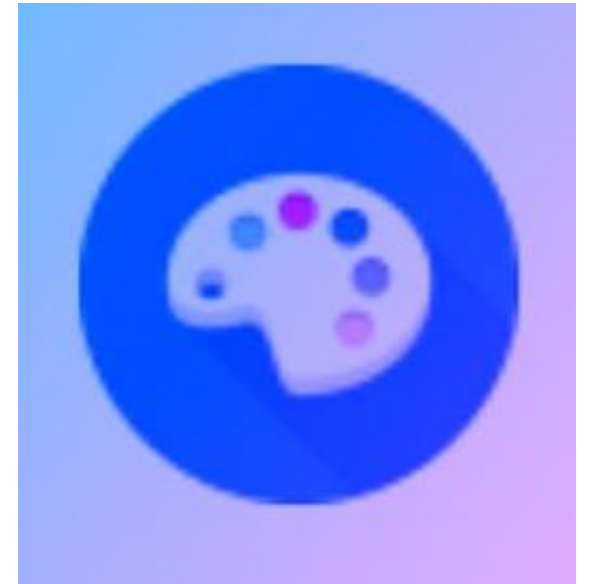
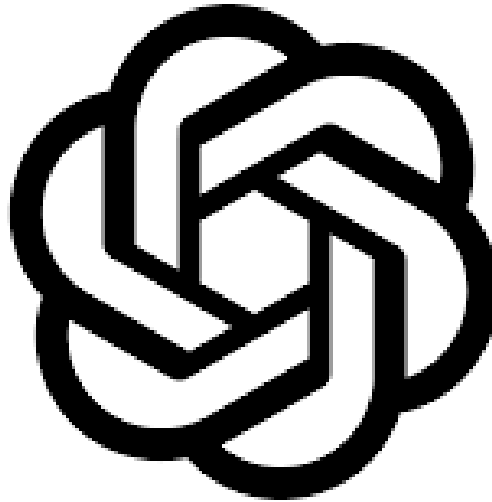
- What's a Diffusion Model and why we should know about it?
- **Diffusion Model and its building blocks**
- Build intuition for **Diffusion Model and its building blocks for any application X**
 - one mental Lego kit for all domains!

Disclaimer – This session will be as devoid of equations and code as possible. A detailed, tailored and guided walkthrough can be scheduled for later depending on interest.



Some popular models out there

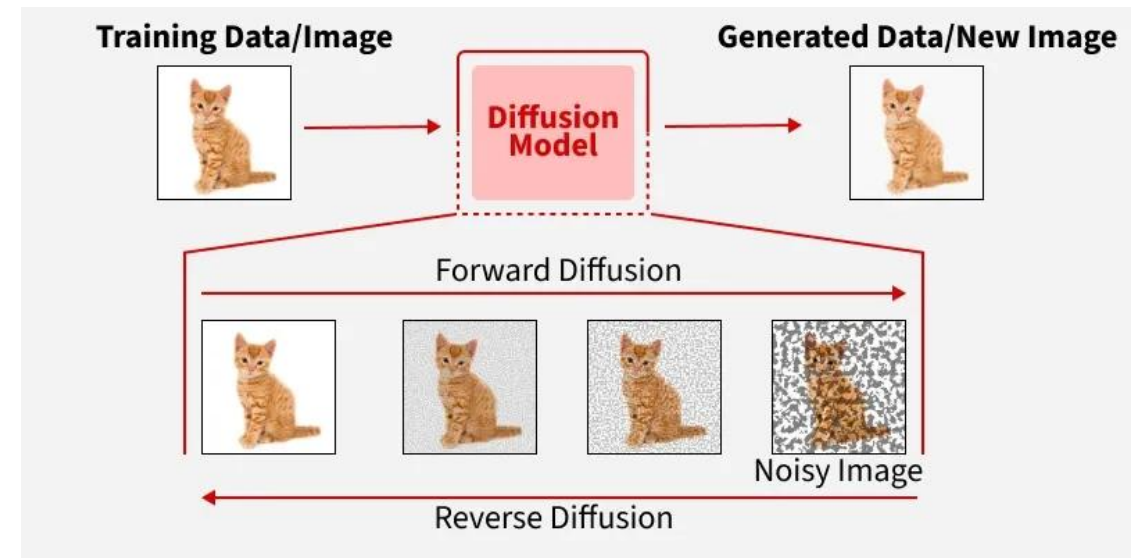
- Everyone has heard about it somewhere!
- Everyone has used it somewhere! Let's see if you can guess these



Intuition behind Diffusion Models

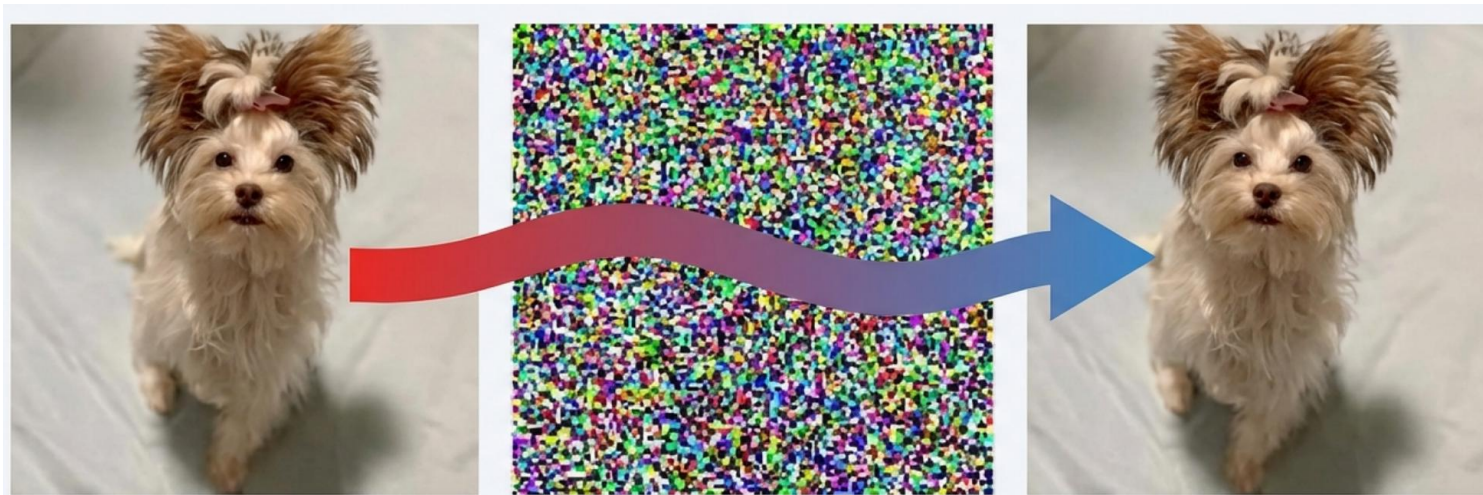
Diffusion in “**Action**”

To train a model to generate an image of a **dragon's eyes**



Why it's important?

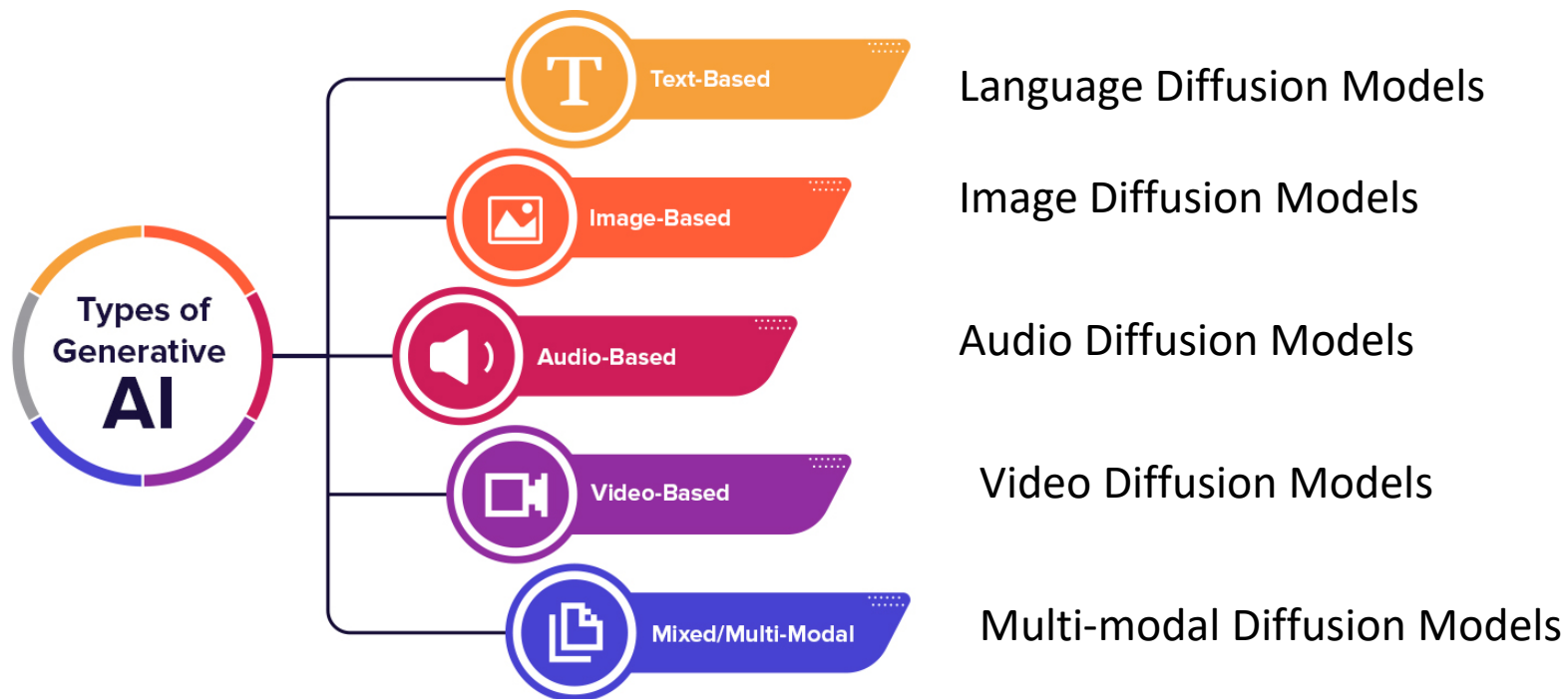
- Model any complex large scale data with ease!
- Any complex data can be represented as numbers with some structure.
 - Diffusion models can help capture this structure and reproduce these numbers in some form



“A central problem in machine learning involves modeling complex data-sets... [Diffusion] simultaneously achieves both flexibility and tractability.” — Sohl-Dickstein et al. (2015)

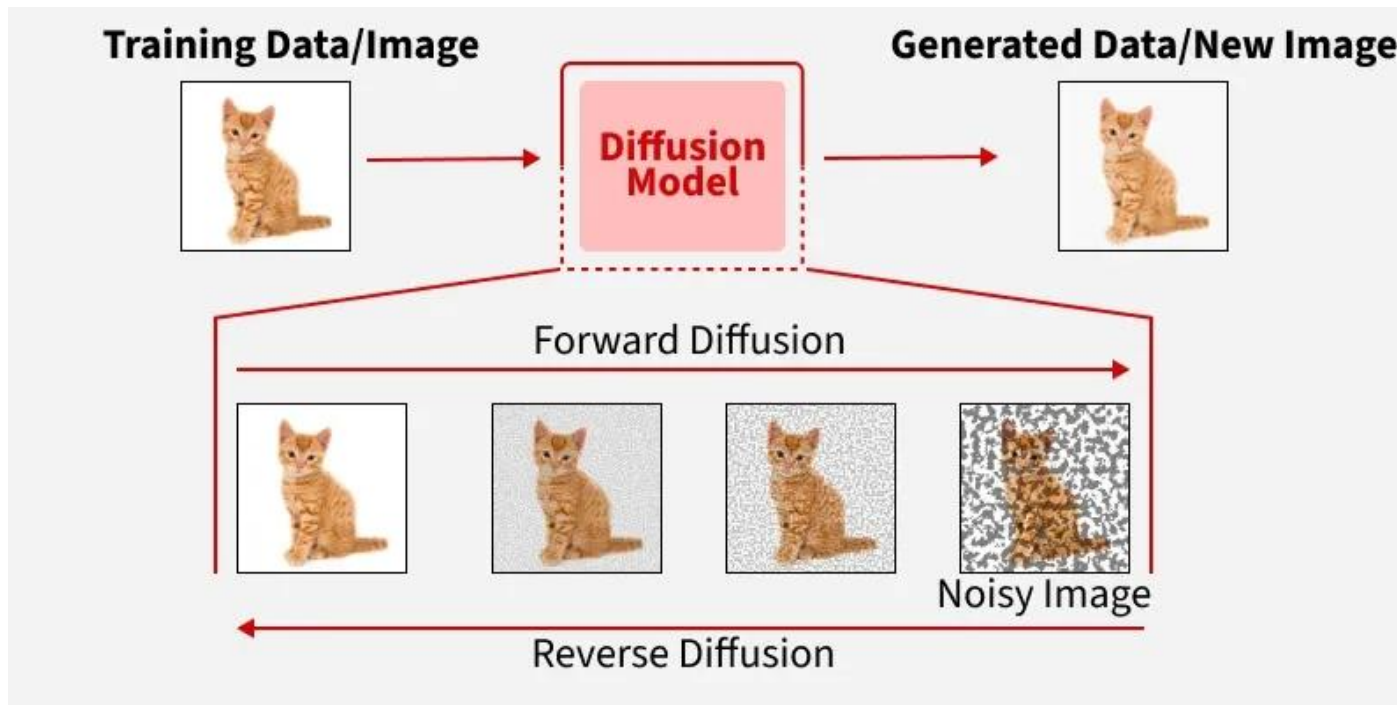
Why it's important?

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What's a diffusion model

- A “generative” model used to generate data given certain constraints.
- Takes noise as input and generates the data.

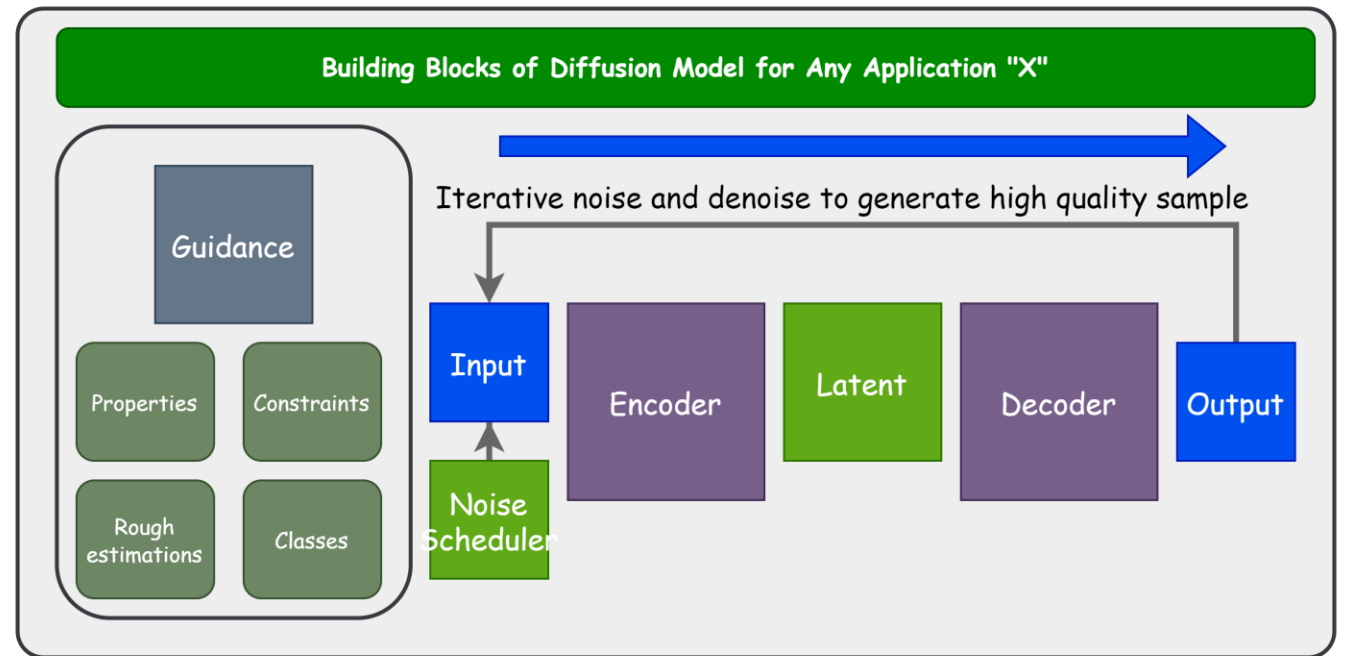


Key Research question behind Diffusion models:

- How can I generate a realistic deepfake for any domain?

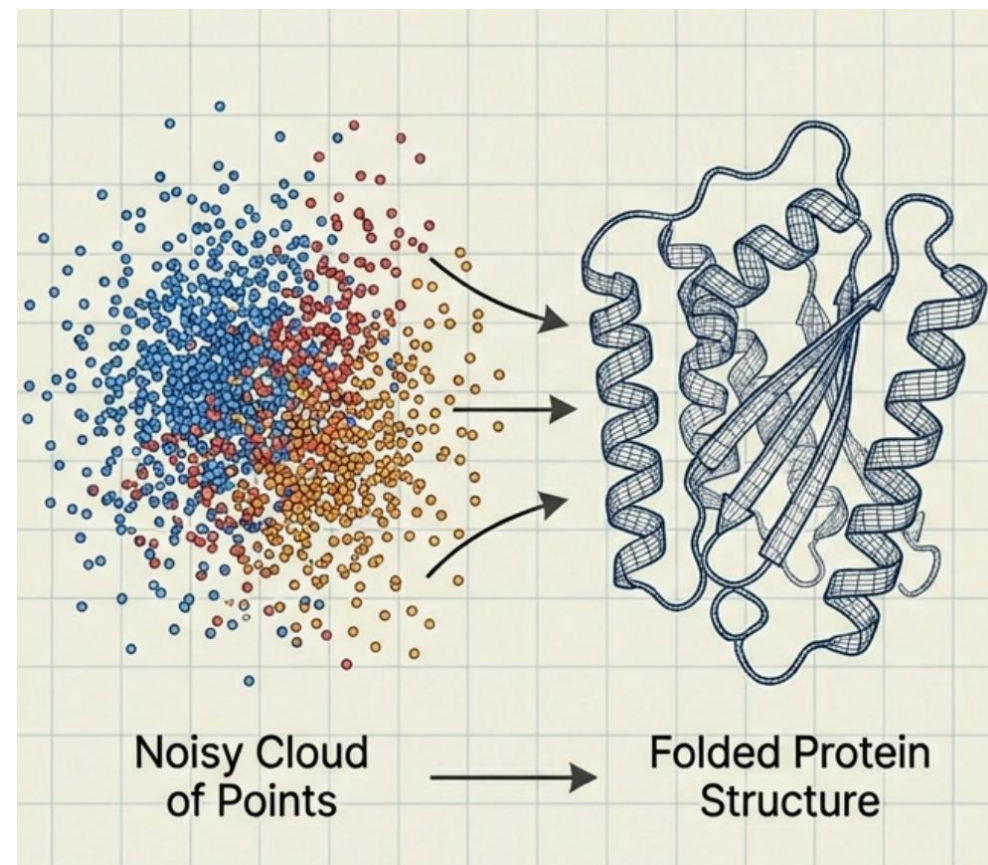
Diffusion models' building blocks

Block	Meaning
Input	Start from noise (Chaos)
Encoder	Understand structure (Understanding)
Embedding	Convert to meaningful numbers (Represent/Sketch)
Conditioning	Tell model what you want (Instructions)
Decoder	Remove noise step-by-step (Refinement)
Output	New data sample (Creation)



Drug Discovery

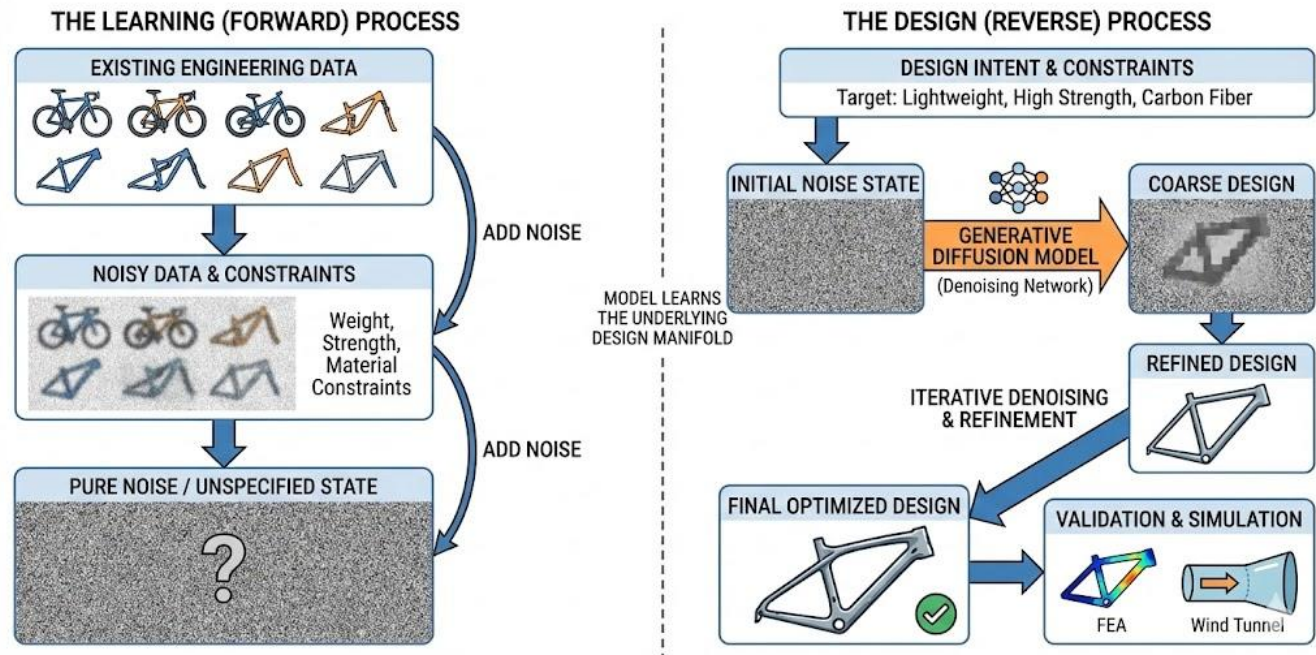
Block	Intuition
Input	Random noisy molecule (scrambled atoms/bonds)
Encoder	Reads molecular structure
Embedding	Converts molecule into numbers
Conditioning / Guidance	Desired property (low toxicity, high binding)
Decoder	Builds molecule step-by-step removing noise
Output	New valid drug candidate molecule



Engineering Design

Block	Intuition
Input	Random shape
Encoder	Reads geometry
Embedding	Shape as vectors
Conditioning / Guidance	Constraints (lightweight, strong)
Decoder	Refines geometry
Output	Optimized design

DIFFUSION MODELS IN ENGINEERING DESIGN: AN INTUITIVE SCHEMATIC



Image/Video Synthesis

Block	Intuition
Input	Static noise image
Encoder	Reads rough shapes
Embedding	Image features as vectors
Conditioning / Guidance	Text prompt (“cat wearing crown”)
Decoder	Sharpens image step-by-step
Output	Final realistic image

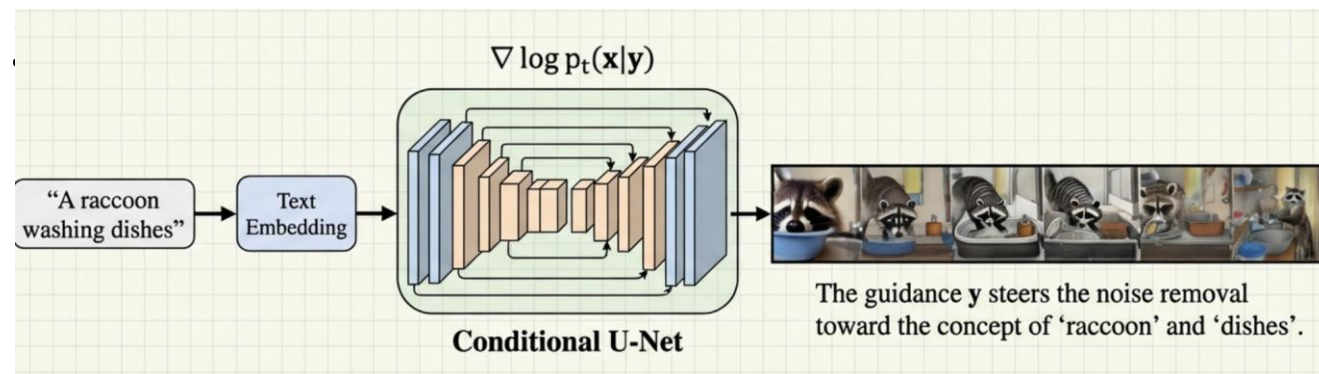
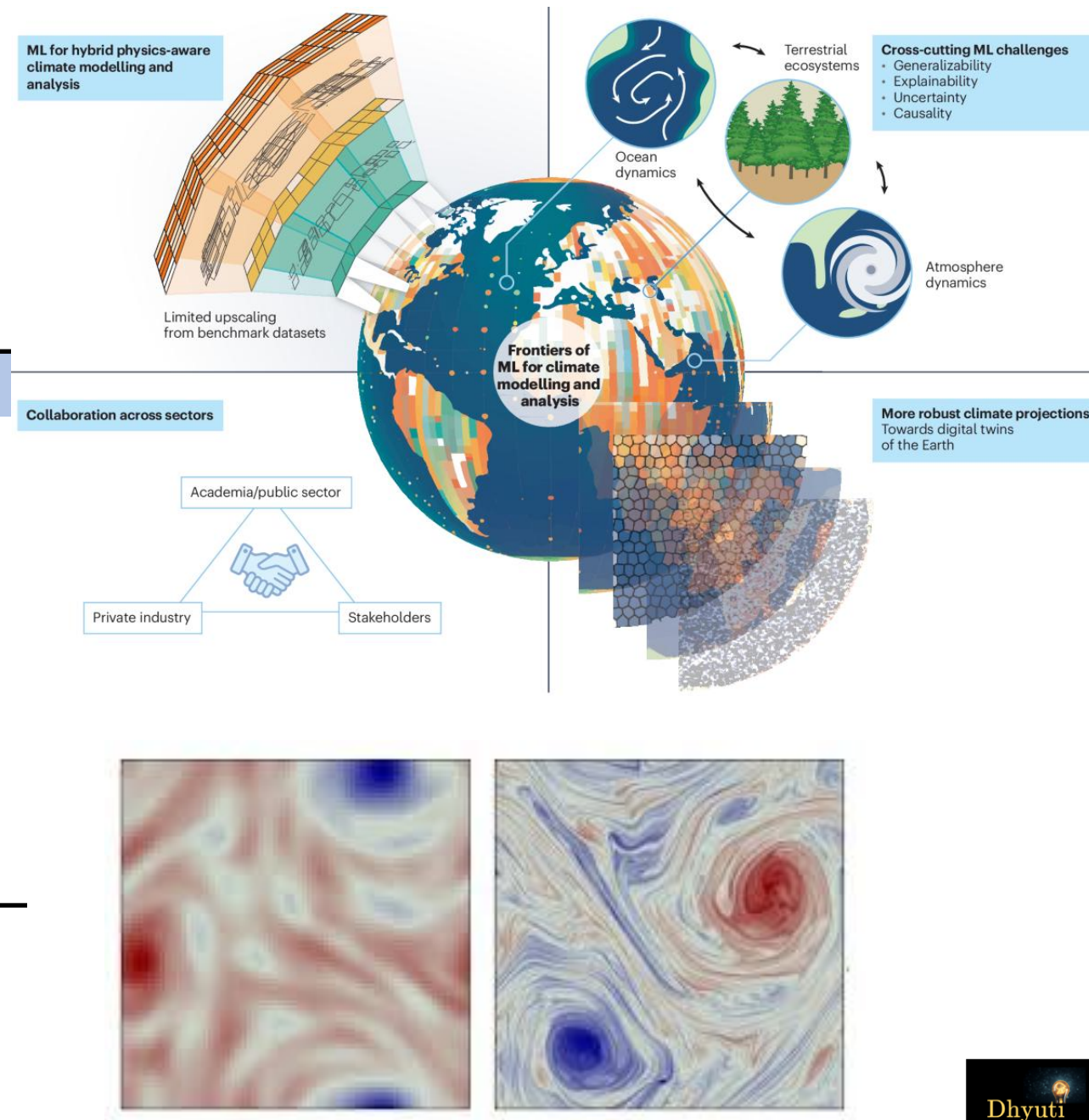


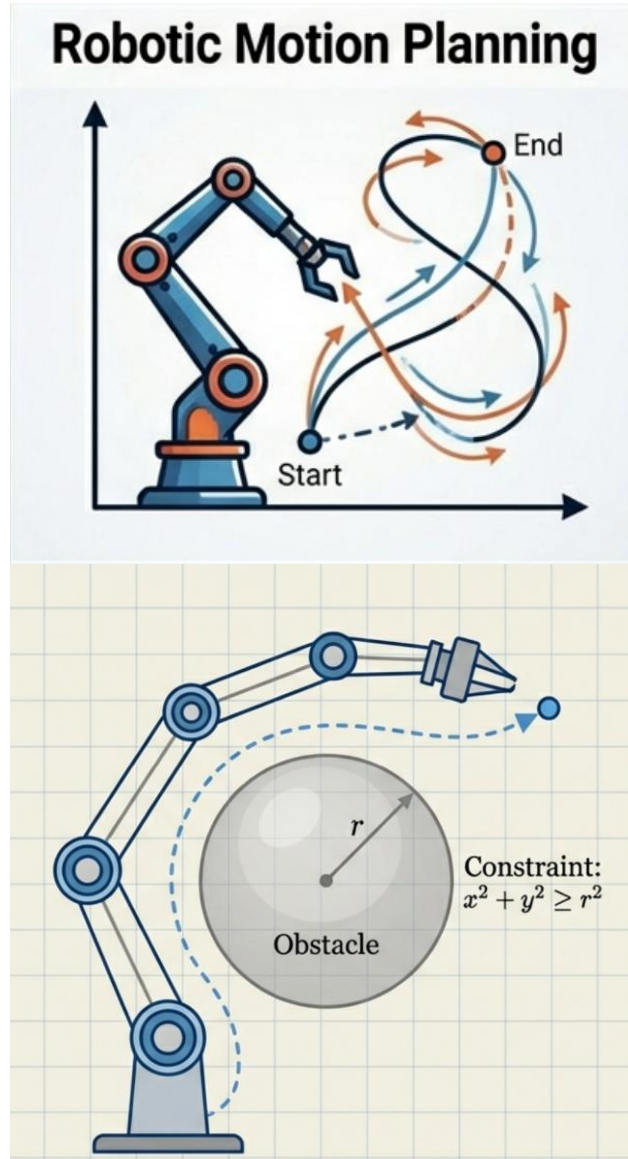
Image Synthesis

Block	Intuition
Input	Noisy climate field
Encoder	Reads spatial weather patterns
Embedding	Climate state vectors
Conditioning / Guidance	CO ₂ levels, season
Decoder	Refines prediction
Output	Climate scenario map



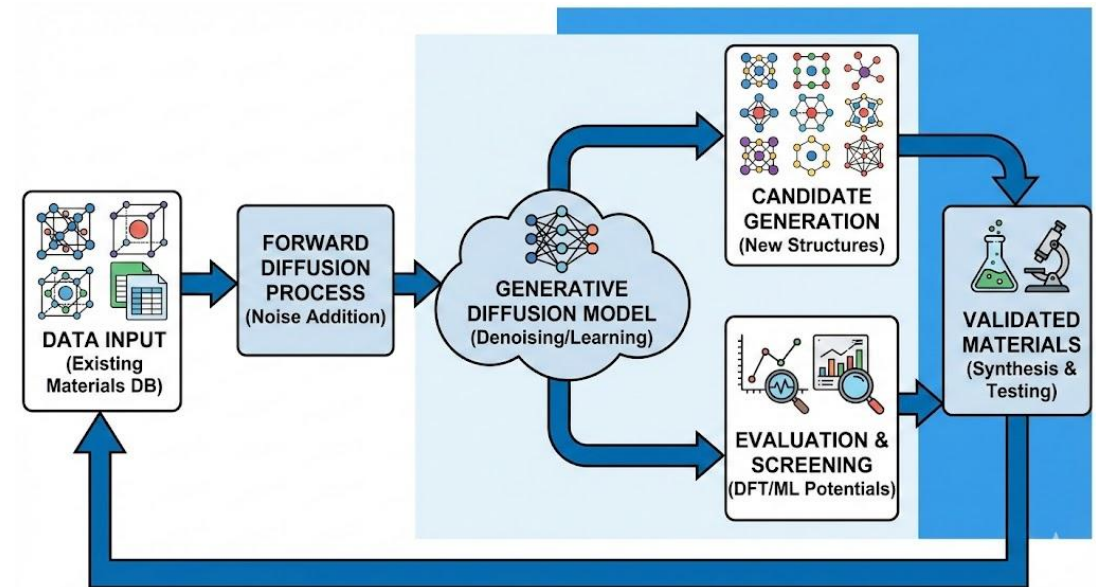
Robotics and Motion Control

Block	Intuition
Input	Random movement plan
Encoder	Understands robot state
Embedding	Motion as numbers
Conditioning / Guidance	Goal (“pick cup”, “avoid obstacle”)
Decoder	Refines motion safely
Output	Smooth robot action sequence



New Material Discovery

Block	Intuition
Input	Random atomic structure
Encoder	Reads crystal layout
Embedding	Atomic features
Conditioning / Guidance	Strength, heat resistance
Decoder	Refines structure
Output	New material design



Explain Diffusion to anyone!

Age Group	Explanation
5-year-old	“The computer starts with messy scribbles and slowly turns them into a nice picture.”
10-year-old	“It learns how to clean noisy pictures step by step until they look real.”
Teenager (15+)	“The model learns patterns from many examples and removes randomness to create new images or ideas.”
College Student	“A diffusion model learns how to reverse noise added to data, generating new samples by denoising step-by-step.”
Adult (non-technical)	“It’s a system that creates realistic content by starting from noise and refining it gradually using learned rules.”
Engineer / Scientist	“A probabilistic generative model that learns the reverse diffusion process to sample from complex data distributions under conditioning.”



Thank you



Let us connect on
LinkedIn!